

NEWINGTON COLLEGE



Trial Examination

12 MATHEMATICS

2003

Extension 1

Time allowed: 2 hours (plus five minutes reading time)

DIRECTIONS TO CANDIDATES

- All questions may be attempted.
- In every question, show all necessary working.
- Marks may not be awarded for careless or badly arranged work.
- Approved silent calculators may be used.
- A table of standard integrals is provided for your convenience.
- The answers to the questions in this paper are to be returned in separate bundles clearly marked Question 1, Question 2, etc.
- Each bundle must show the candidate's computer number.
- The questions are not necessarily arranged in order of difficulty. Candidates are advised to read the whole paper carefully at the start of the examination.
- Unless otherwise stated, candidates should leave their answers in simplest exact form.

Page 1

Question 1 12 marks

marks

a) Differentiate $\tan^{-1} \frac{x}{3}$.

2

b) Evaluate:

6

(i) $\int_1^{\sqrt{5}} \frac{x}{\sqrt{4-x^2}} dx$ using the substitution $u = 4 - x^2$.

(ii) $\int_0^1 \sqrt{1-x^2} dx$ using the substitution $x = \sin \theta$.

c) Solve the equation $3\sin \theta + 4\cos \theta = 2.5$ for values of θ between 0° and 360° .
Give your answer correct to the nearest minute. 4**Question 2** 12 marks **Start a New Booklet**

a) (i) Show that $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{dv}{dt}$.

5

(ii) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$ where x metres is the displacement from the origin. Initially, the particle is at the origin with velocity 2 ms^{-1} .
Prove that $v = 2e^{\frac{-x}{2}}$.(iii) What happens to v as x increases without bound?b) (i) By considering the graph of $y = e^x$, show that the equation $e^x + x + 1 = 0$ has only one real root and that this root is negative. 4(ii) Taking $x = -1.5$ as a first approximation to this root, use one application of Newton's method to find a better approximation.c) In how many ways can the letters of the word *GEOMETRY* be arranged in a straight line if the vowels must occupy the 2nd, 4th and 6th places.
(NOTE: The vowels in the English alphabet are the letters A, E, I, O, U). 3

Page 2

Question 3 12 marks Start a New Booklet**marks**

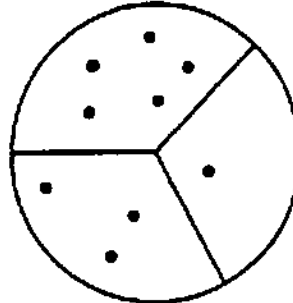
- a) Find the general solution for $\sqrt{3} \sin 2\theta = \cos 2\theta$. 3
- b) The region bounded by the curve $y = \sin x$, the x -axis and the lines 3
 $x = \frac{\pi}{12}$ and $x = \frac{\pi}{4}$ is rotated through one complete revolution about the x -axis.
 Find the volume of the solid so formed.
- c) Two points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$. 6
 (i) Show that the equation of the tangent to the parabola at P is $y = px - p^2$.
 (ii) The tangent at P and the line through Q parallel to the y axis intersect at T . Find the coordinates of T .
 (iii) Write down the coordinates of M , the midpoint of PT .
 (iv) Determine the locus of M when $pq = -1$.

Question 4 12 marks Start a New Booklet

- a) If $\tan A$ and $\tan B$ are the roots of the equation $3x^2 - 5x - 1 = 0$, find the value 3
 of $\tan(A + B)$.
- b) A particle is moving with simple harmonic motion. When it is at a distance d 5
 from the centre of motion, its speed is V . If its speed is $\frac{V}{2}$ when the distance
 from the centre is $2d$, show that the period of the motion is $\frac{4\pi d}{V}$ and the
 amplitude is $d\sqrt{5}$.
- c) The rate at which a body cools in air is assumed to be proportional to the 4
 difference between its temperature T and the constant temperature S of the
 surrounding air. This can be expressed by the differential equation
 $\frac{dT}{dt} = k(T - S)$ where t is the time in hours and k is a constant.
- (i) Show that $T = S + Be^{kt}$, where B is a constant, is a solution of the
 differential equation.
- (ii) A heated body cools from 80°C to 40°C in 2 hours. The air
 temperature S around the body is 20°C . Find the temperature of the
 body after one further hour has elapsed. Give your answer correct to the
 nearest degree.

Question 5 12 marks Start a New Booklet**marks**

- a) Nine points lie inside a circle. No three of the points are collinear. Five of the points lie in sector 1, three lie in sector 2, and the other point lies in sector 3. 5



- (i) Show that 84 triangles can be made using these points as vertices.
 - (ii) One triangle is chosen at random from all the possible triangles. Find the probability that the vertices of the triangle chosen lie one in each sector.
 - (iii) Find the probability that the vertices of the triangle chosen lie all in the same sector.
- b) Find the roots of the equation $x^3 - 12x^2 + 12x + 80 = 0$ given that they are three consecutive terms in an Arithmetic Series. 3
- c) Consider the binomial expansion $1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n$. 4

(i) Show that $1 - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$.

(ii) Show that $1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + (-1)^n \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}$.

Question 6 12 marks Start a New Booklet

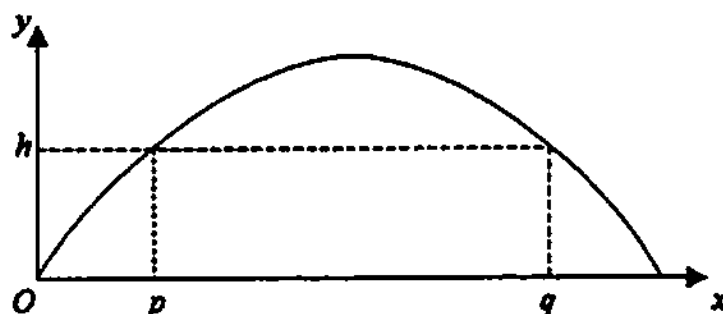
- a) Colour-blindness affects 5% of all men. What is the probability that any random sample of 20 men should contain:
- (i) no colour-blind men.
 - (ii) only one colour-blind man.
 - (iii) two or more colour-blind men.

- b) When $(3 + 2x)^n$ is expanded in increasing powers of x , it is found that the coefficients of x^3 and x^6 have the same value. Find the value of n and show that the two coefficients mentioned are greater than all other coefficients in the expansion. 7

Question 7 12 marks Start a New Booklet

- a) Prove by induction that $2^3 + 4^3 + 6^3 + \dots + (2n)^3 = 2n^2(n+1)^2$. 5

- b) 7



A particle is projected with velocity $V \text{ ms}^{-1}$ from a point O at an angle of elevation α . Axes Ox and Oy are taken horizontally and vertically through O . The particle just clears two vertical chimneys of height h meters at horizontal distances of p metres and q metres from O . The acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.

- (i) Write down expressions for the horizontal displacement x and the vertical displacement y of the particle after time t seconds.
- (ii) Show that $V^2 = \frac{5p^2(1 + \tan^2 \alpha)}{p \tan \alpha - h}$.
- (iii) Show that $\tan \alpha = \frac{h(p + q)}{pq}$.

END OF PAPER

Solutions

①

YA 12.1 RIN2

EXT 1

$$1/(a) \quad \frac{\frac{1}{3}}{1 + \frac{x^2}{9}} = \frac{3}{x^2 + 9}$$

$$(4)(i) \int_1^{\sqrt{3}} \frac{x}{\sqrt{4-x^2}} dx$$

$$\begin{aligned} u &= 4 - x^2 \\ \frac{du}{dx} &= -2x \\ x dx &= -\frac{du}{2} \end{aligned}$$

$$x=1, u=3$$

$$x=\sqrt{3}, u=2$$

$$= - \int_3^2 \frac{du}{2\sqrt{u}}$$

$$= - \int_3^2 \frac{1}{2} u^{-\frac{1}{2}} du$$

$$= \left[u^{\frac{1}{2}} \right]_2^3$$

$$= \sqrt{3} - \sqrt{2}$$

$$(ii) \int_0^1 \sqrt{1-x^2} dx$$

$$\begin{aligned} x &= \sin \theta \\ dx &= \cos \theta d\theta \end{aligned}$$

$$x=0, \theta=0$$

$$x=1, \theta=\frac{\pi}{2}$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2 \theta} \cdot \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} + 0 - (0+0) \right)$$

(2)

$$c) 5\left(\frac{3}{5}\sin\theta + \frac{4}{5}\cos\theta\right) = 2.5$$

$$\sin(\theta + \alpha) = \frac{1}{2}, \quad \alpha = \sin^{-1}\frac{4}{5}$$

$$\theta + \alpha = 30^\circ, 150^\circ, 390^\circ$$

$$\theta = 30^\circ - \alpha, 150^\circ - \alpha, 390^\circ - \alpha$$

$$= -23^\circ 52', 96^\circ 52', 336^\circ 52'$$

$$= 96^\circ 52', 336^\circ 52' \text{ (nearest minute)}$$

2/

$$\begin{aligned} (a) (i) \quad \frac{dv}{dt} &= \frac{dv}{dx} \cdot \frac{dx}{dt} \\ &= v \frac{dv}{dx} \\ &= \frac{d}{dv}\left(\frac{1}{2}v^2\right) \cdot \frac{dv}{dx} \\ &= \frac{d}{dx}\left(\frac{1}{2}v^2\right) \end{aligned}$$

$$(ii) \quad \ddot{x} = -2e^{-x}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -2e^{-x}$$

$$\frac{1}{2}v^2 = 2e^{-x} + C$$

$$x=0, v=2$$

$$2 = 2e^0 + C, \quad C=0$$

$$\frac{1}{2}v^2 = 2e^{-x}$$

$$v^2 = 4e^{-x}$$

$$v = \pm 2e^{-\frac{x}{2}}$$

Initially $v > 0$ and $v^2 \neq 0 \therefore$ reject $-ve \quad v$

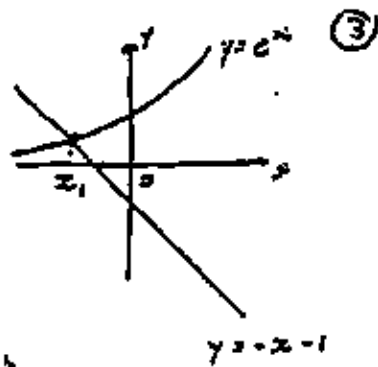
$$v = 2e^{-\frac{x}{2}}$$

(iii) as $x \rightarrow \infty, v \rightarrow 0$

b)

$$y = e^x$$

$$y = -x - 1$$



$$f(x) = e^x + x + 1$$

$$f'(x) = e^x + 1$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= -1.5 - \frac{e^{-1.5} - 1.5 + 1}{e^{-1.5} + 1}$$

$$= -1.27 \quad (\text{correct to 2 dec. pl})$$

$$5) \quad _ \vee _ \vee _ \vee _ _$$

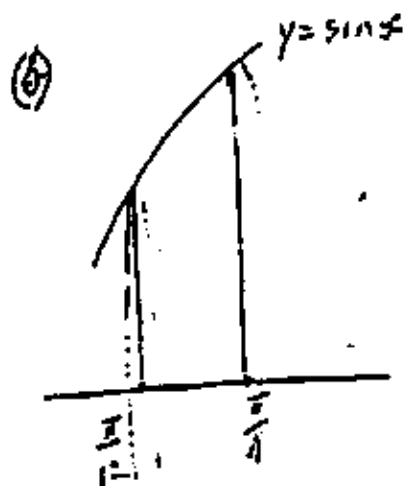
$$\begin{aligned} \text{No. of arrangements} &= \frac{3! \cdot 5!}{2!} \\ &= 360 \end{aligned}$$

3(a)

$$\Rightarrow \tan 2\theta = \frac{1}{\sqrt{3}}$$

$$2\theta = n\pi + \frac{\pi}{6}$$

$$\theta = n\frac{\pi}{2} + \frac{\pi}{12}$$



$$V = \pi \int_{\pi/12}^{\pi/4} \sin^2 x \, dx$$

$$= \frac{\pi}{2} \int_{\pi/12}^{\pi/4} (1 - \cos 2x) \, dx$$

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\pi/12}^{\pi/4}$$

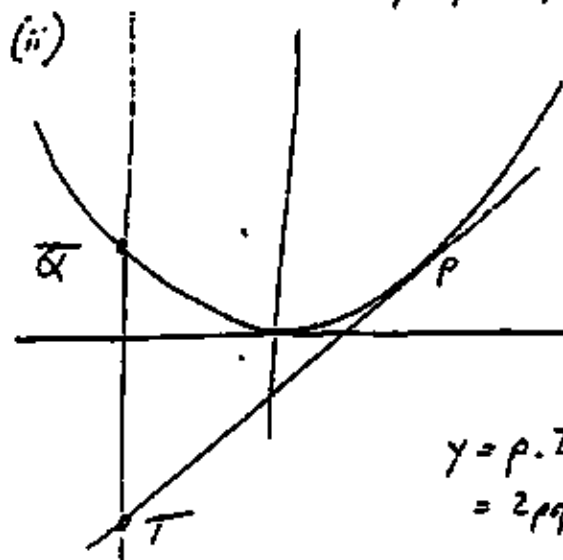
$$= \frac{\pi}{2} \left(\frac{\pi}{4} - \frac{1}{2} - \left(\frac{\pi}{12} - \frac{1}{2} - \frac{1}{2} \right) \right)$$

$$= \frac{\pi}{2} \left(\frac{\pi}{6} - \frac{1}{4} \right)$$

$$= \frac{\pi(2\pi - 3)}{24} \text{ units}^3$$

(c) (i) $\frac{dy}{dx} = \frac{x}{2y}$ at $P(2p, p^2)$
 $= p$

required equation: $y - p^2 = p(x - 2p)$
 $= px - 2p^2$
 $y = px - p^2$



$$y = p \cdot 2q - p^2$$

$$= 2pq - p^2$$

$$\therefore T(2a, 2pq - p^2)$$

3/ cont.

(5)

(c)

iii) $M(p+q, pq)$

iv) $y = -1$

4/ (a) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$\tan A + \tan B = \frac{5}{3}$$

$$\tan A \tan B = -\frac{1}{3}$$

$$\therefore \tan(A+B) = \frac{\frac{5}{3}}{1 - (-\frac{1}{3})}$$
$$= \frac{5}{4}$$

(b) $\ddot{x} = -n^2 x$

$$\frac{1}{2} v^2 = -\frac{n^2 x^2}{2} + C$$

$v = V, x = d$

$$\therefore C = \frac{1}{2} V^2 + \frac{n^2 d^2}{2}$$

$$\frac{1}{2} v^2 = -\frac{n^2 x^2}{2} + \frac{1}{2} V^2 + \frac{n^2 d^2}{2}$$

$$v^2 = V^2 + n^2 (d^2 - x^2)$$

$v = \frac{V}{2}, x = 2d$

$$\left(\frac{V}{2}\right)^2 = V^2 + n^2 (d^2 - 4d^2)$$

$$n^2 = \left(V^2 - \frac{V^2}{4}\right) \div 3d^2$$

$$= \frac{3V^2}{4} \times \frac{1}{3d^2}$$

$$= \underline{V^2}$$

(b)

$$\begin{aligned}
 \text{Period} &= \frac{2\pi}{n} \\
 &= 2\pi \cdot \frac{2d}{\sqrt{v}} \\
 &= \frac{4\pi d}{\sqrt{v}}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } v=0, \quad v^2 + n^2(d^2 - x^2) &= 0 \\
 v^2 + \frac{v^2}{4d^2}(d^2 - x^2) &= 0 \\
 \therefore x^2 - d^2 &= v^2 \cdot \frac{4d^2}{v^2}
 \end{aligned}$$

$$\begin{aligned}
 x^2 &= 5d^2 \\
 \therefore \text{amplitude} &= \sqrt{5d^2} \\
 &= d\sqrt{5}
 \end{aligned}$$

(c)

$$\begin{aligned}
 \text{(i)} \quad \frac{dT}{dt} &= kBe^{kt} \\
 &= k(s + Be^{kt} - s) \\
 &= k(T - s)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad t=0, T &= 80^\circ \\
 80^\circ &= 20^\circ + Be^0 \\
 B &= 60 \\
 T &= 20 + 60e^{kt} \\
 40 &= 20 + 60e^{2k} \\
 e^{2k} &= \frac{1}{3} \\
 2k &= \ln\left(\frac{1}{3}\right) \\
 k &= \frac{1}{2} \ln\left(\frac{1}{3}\right)
 \end{aligned}$$

$$\begin{aligned}
 t=3, T &= 20 + 60e^{\frac{1}{2} \ln\left(\frac{1}{3}\right) \cdot 3} \\
 &= 20 + 60 \ln\left(\frac{1}{3}\right)^{3/2} \\
 &= 20 + 60 \cdot 3^{-3/2}
 \end{aligned}$$

(7)

$$5/(a) \quad (i) {}^9C_3 = 84$$

$$(ii) \frac{{}^5C_1 {}^3C_1}{84} = \frac{15}{84}$$

$$= \frac{5}{28}$$

$$(iii) \frac{{}^9C_3 + 1}{84} = \frac{11}{84}$$

$$(b) x^3 - 12x^2 + 12x + 80 = 0$$

Let roots be $x-m, x, x+m$

$$\text{sum of roots: } 3d = 12$$

$$d = 4$$

$$\text{product of roots: } (4-m)4(4+m) = 80$$

$$16 - m^2 = 20$$

$$m = \pm 2$$

$\therefore$ roots are $-2, 4, 10$

$$(c) 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n$$

$$(i) x = -1, \quad 1 - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$$

(ii) Integrate both sides wrt x

$$x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \dots + \frac{1}{n+1}\binom{n}{n}x^{n+1} = \frac{1}{n+1}(1+x)^{n+1} + C$$

$$\text{Let } x = 0 \quad \therefore \frac{1}{n+1} + C = 0, \quad C = -\frac{1}{n+1}$$

$$x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \dots + \frac{1}{n+1}\binom{n}{n}x^{n+1} = \frac{1}{n+1}(1+x)^{n+1} - \frac{1}{n+1}$$

$$\text{Let } x = -1, \quad -1 + \frac{1}{2}\binom{n}{1} - \frac{1}{3}\binom{n}{2} + \dots + (-1)^{n+1} \frac{1}{n+1} \binom{n}{n} = -\frac{1}{n+1}$$

$$1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + (-1)^n \frac{1}{n+1} \binom{n}{n} = \frac{1}{n+1}$$

(8)

$$(a) \quad p = \frac{1}{20}, \quad q = \frac{19}{20}$$

$$(i) \quad (q+p)^{20} = \sum_{r=0}^{20} \binom{20}{r} q^{20-r} p^r$$

$$P(\text{no colour-blind}) = \binom{20}{0} \left(\frac{19}{20}\right)^{20}$$

$$= \left(\frac{19}{20}\right)^{20}$$

$$(ii) \quad P(\text{one colour-blind}) = \binom{20}{1} q^{19} p$$

$$= 20 \cdot \left(\frac{19}{20}\right)^{19} \cdot \frac{1}{20}$$

$$= \left(\frac{19}{20}\right)^{19}$$

$$(iii) \quad P(\text{at least 2 colour-blind}) \\ = 1 - \left(\left(\frac{19}{20}\right)^{20} + \left(\frac{19}{20}\right)^{19}\right)$$

$$(b) \quad (3+2x)^n = \sum_{r=0}^n \binom{n}{r} 3^{n-r} (2x)^r$$

$$\binom{n}{5} 3^{n-5} (2x)^5 = \binom{n}{6} 3^{n-6} (2x)^6$$

$$\binom{n}{5} \div \binom{n}{6} = \frac{3^{n-6}}{3^{n-5}} \cdot \frac{2^6}{2^5}$$

$$\frac{n!}{(n-5)! 5!} \cdot \frac{(n-6)! 6!}{n!} = \frac{2}{3}$$

$$\frac{6}{n-5} = \frac{2}{3}$$

$$2n - 10 = 18$$

$$n = 14$$

6/6) cont

(9)

$$\frac{t_{r+1}}{t_r} = \frac{{}^{14}C_r 3^{14-r} (2x)^r}{{}^{14}C_{r-1} 3^{15-r} (2x)^{r-1}}$$

where t_r = r th term in the expansion

$$= \frac{14!}{(14-r)! r!} \cdot \frac{(15-r)! (r-1)!}{14!} \cdot \frac{2x}{3}$$

Let c_r = coefficient of the r th term

$$\frac{c_{r+1}}{c_r} = \frac{15-r}{r} \cdot \frac{2}{3}$$

$$= \frac{30-2r}{3r}$$

$$\frac{c_{r+1}}{c_r} = 1 \quad \text{when } r=6 \quad \therefore c_6 = c_7$$

$$\frac{c_{r+1}}{c_r} > 1 \quad \text{when } r < 6 \quad \therefore c_1 < c_2 < c_3 < c_4 < c_5 < c_6$$

$$\frac{c_{r+1}}{c_r} < 1 \quad \text{when } r > 6 \quad \therefore c_7 > c_8 > c_9 > c_{10} > c_{11} > c_{12} < c_{13}$$

c_6 & c_7 are greatest coefficients

7/1a) Step 1 Let $n=1$

$$\text{LHS} = (2 \cdot 1)^3$$

$$= 8$$

$\therefore$ true for $n=1$

$$\text{RHS} = 2 \cdot 1^2 (1+1)^2$$

$$= 8$$

Step 2 Assume result true for $n=k$, k is a positive integer

$$\text{i.e. } 2^3 + 4^3 + 6^3 + \dots + (2k)^3 = 2k^2(k+1)^2$$

Step 3 Prove result true for $n=k+1$

$$\therefore \text{prove } 2^3 + 4^3 + 6^3 + \dots + (2k)^3 + (2(k+1))^3 = 2(k+1)^2((k+1)+1)^2$$

7(a) cont.

(10)

$$\begin{aligned} \text{LHS} &= 2k^2(k+1)^2 + (2(k+1))^3 \quad \text{from assumption} \\ &= 2(k+1)^2(k^2 + 4(k+1)) \\ &= 2(k+1)^2(k^2 + 4k + 4) \\ &= 2(k+1)^2(k+2)^2 \\ &= \text{RHS} \end{aligned}$$

Step 4 Result is true for $n=1$. Hence it is true for $n=1+1=2$, $n=2+1=3$ etc. $\therefore$ The result is true for all positive integers

(b) (i) $x = vt \cos \alpha$
 $y = vt \sin \alpha - \frac{1}{2} g t^2$

(ii) $p = vt \cos \alpha$
 $t = \frac{p}{v \cos \alpha}$

$$h = v \cdot \frac{p}{v \cos \alpha} \cdot \sin \alpha - \frac{1}{2} g \frac{p^2}{v^2 \cos^2 \alpha}$$

$$= p \tan \alpha - \frac{5p^2}{v^2 \cos^2 \alpha}$$

$$= p \tan \alpha - \frac{5p^2}{v^2} (\tan^2 \alpha + 1)$$

$$5p^2 (\tan^2 \alpha + 1) = v^2 (p \tan \alpha - h)$$

$$v^2 = \frac{5p^2 (\tan^2 \alpha + 1)}{p \tan \alpha - h}$$

(iii) $v^2 = \frac{5q^2 (\tan^2 \alpha + 1)}{q \tan \alpha - h}$

$$\therefore \frac{5p^2 (\tan^2 \alpha + 1)}{p \tan \alpha - h} = \frac{5q^2 (\tan^2 \alpha + 1)}{q \tan \alpha - h}$$

$$p^2 (q \tan \alpha - h) = q^2 (p \tan \alpha - h)$$

$$\tan \alpha (p^2 q - q^2 p) = p^2 h - q^2 h$$

$$\tan \alpha = \frac{h(p-q)(p+q)}{pq(p-q)} = \frac{h(p+q)}{pq}$$